Scatter Plots — Direction, Form, and Strength
Answer Key

Grade 8 / Pre-Algebra

Quick Answers

1. —	5. 6, 5	9. 34.125, 37.57
2. —	6. 7	10. —
3. 4	7. 2, 15	
4. 1, -6	8. 33	

Curriculum

Level: Grade 8 / Pre-Algebra
8.SP.A.1 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*
Difficulty 1–5 of 5 across 14 tagged checks.

Common wrong answers

6. *If they got 1: read off the smallest x-value instead of the x-value where the smallest y-value sits*

How this sheet is built: students can usually say “positive” or “negative” from a glance and cannot say what makes them sure. So every problem asks for a *number* first and a *description* second — a comparison, an extreme value, a residual, a range — and the description has to cite that number. Grade the citation, not the adjective. A student who writes “strong” with no reference to how close the points sit to the line has produced the same answer a guess produces.

1. Plot A — direction from the two end points.

(a) The leftmost point is at $x = 1$ with $y = 52$; the rightmost is at $x = 8$ with $y = 88$. Comparing them:

$$\boxed{52 < 88}$$

(b) The direction is **positive**. The comparison in (a) says the right-hand end of the plot is higher than the left-hand end, so y tends to increase as x increases. Every point in between continues the climb, which is what makes the comparison representative rather than a coincidence of the two ends.
Grading note: accept “as x goes up, y goes up” as the reason. Do not accept “it goes up to the right” with no numbers — (a) exists so the claim has evidence.

2. Plot B — the other direction.

(a) Leftmost point $(0, 46)$, rightmost point $(35, 16)$:

$$\boxed{46 > 16}$$

(b) The direction is **negative**: as x increases, y tends to **decrease**.
Both plots so far are strongly linear, and the only thing that differs is the sign of the change. That is exactly what "direction" names — nothing else.

3. Plot D — reading the form.

(a) Scan the heights, not the x-values. The lowest point on Plot D is $y = 6$, and it sits above $x = \boxed{4}$.
(b) From left to right the points *fall* from 30 down to 6, and then *rise* again to 33. A straight line cannot do that: a line goes one way only. So the form is **nonlinear** — here, a U-shaped (curved) pattern.
Why the turning point is the evidence: the x-value where the pattern reverses is the thing a line does not have. Naming it is a more convincing answer than "it looks curvy".

4. Residuals: how far one point sits from its trend.

(a) Plot A, at $x = 5$: the line predicts $5(5) + 47 = 72$ and the point is at 73. Residual is actual minus predicted, in that order:

$$73 - 72$$

The Plot A residual is $\boxed{1}$.
(b) Plot E, at $x = 5$: the line predicts $5(5) + 25 = 50$ and the point is at 44:

$$44 - 50$$

The Plot E residual is $\boxed{-6}$.
(c) The Plot A point is closer to its line. Its residual is one unit from zero, while the Plot E residual is six units from zero — and it is the *distance* from zero that matters, not the sign. The negative sign on the Plot E residual says only that the point sits *below* its line.
Watch for: predicted minus actual, which flips both signs. It gives the same distances, so (c) still comes out right, but the sign then means the opposite of what the student says it means. Insist on actual − predicted.

5. Plot C — what "no association" looks like.

(a) The highest y-value on Plot C is 63, sitting above $x = \boxed{6}$.
(b) The lowest is 41, sitting above $x = \boxed{5}$.
(c) On Plot A the highest and lowest values sat at the two *ends* — lowest on the far left, highest on the far right — which is what a direction is. Here the two extremes sit side by side in the middle of the plot, at $x = 5$ and $x = 6$. Knowing x tells you nothing useful about y: the points zig-zag between about 41 and 63 all the way across. That is **no association**.
The habit worth building: "no association" is a description, not a failure to find one. A student should be able to say what would have to be true for there to be a direction, and then point out that it isn't.

6. Plot F — finding the outlier.

(a) The lowest y-value on Plot F is 10, and it sits above $x = \boxed{7}$.

Watch for: answering 1. That is the smallest x-value, not the location of the smallest y-value — the question asks where the lowest *height* is. The point at $x = 1$ has $y = 20$, which is twice the height of the point at $x = 7$.

(b) Cover $(7, 10)$ and the remaining seven points read $20, 26, 31, 37, 42, 48, 59$ — climbing by roughly 5 or 6 each step. Direction: **positive**. Form: **linear**. Strength: **strong** (the points sit almost exactly on a line).

(c) The outlier makes the association *look* weaker than it is. One point sitting 35 or so below the pattern spoils the visual impression of a tight line, even though seven of the eight points are nearly perfectly linear. The honest description of Plot F is "strong positive linear association with one outlier at $x = 7$" — and that sentence is more useful than either "strong" or "weak" alone.

7. Putting a number on strength.

(a) Range means largest value minus smallest value.

Plot A residuals run from -1 to 1:

$$1 - (-1)$$

so the range is $\boxed{2}$.

Plot E residuals run from -7 to 8:

$$8 - (-7)$$

so the range is $\boxed{15}$.

(b) Plot A is stronger. Its residual range of 2 means every one of its eight points sits within one unit of the trend line; Plot E's range of 15 means its points wander up to eight units either side.

What the range of the residuals measures: the width of the band the points occupy around the line — how thick the cloud is. That is precisely what "strength" means. It is not the slope: both plots here have the same trend slope of 5, and they differ enormously in strength. Slope answers "how fast", strength answers "how reliably".

This is the single most valuable idea on the sheet, and the one problem 10 comes back to.

8. Plot D — why "direction" is the wrong question.

(a) The highest y-value anywhere on Plot D is $\boxed{33}$, at $x = 8$.

(b) The student's evidence is correct: the last point (33) really is higher than the first (30). The conclusion is still wrong, and the reason is **form**.

"Positive association" means y tends to increase *throughout* — it is a statement about the whole plot, not about its two end points. Plot D's form is a curve: y falls for the first half and rises for the second. Comparing only the ends hides everything in between, and here the two ends happen to be nearly the same height (30 and 33) with a value of 6 buried in the middle.

The rule to state: check the form *before* you name a direction. On a nonlinear plot, "positive" and "negative" do not describe anything — the right description is "decreasing then increasing".

9. What one outlier does to a summary.

(a) Add all eight y-values and divide by 8:

$$20 + 26 + 31 + 37 + 42 + 48 + 10 + 59 = 273 \qquad \frac{273}{8}$$

The mean is $\boxed{34.125}$.

(b) Drop the outlier 10 and divide the remaining total by 7:

$$273 - 10 = 263 \qquad \frac{263}{7} = 37.5714\ldots$$

Rounded to the nearest hundredth, $\boxed{37.57}$.

(c) Removing one value out of eight moved the mean by about 3.4 — from 34.13 up to 37.57. That is a large move for a single point, and it happens because the mean adds every value in with equal weight, so a value far from the rest pulls hard.

What it shows: in a small data set, one unusual point can drive a summary. Before quoting a mean, look at the plot and ask whether one point is doing the work. This is the numerical version of what problem 6(c) said in words.

Watch for: subtracting the outlier from the total but still dividing by 8, which gives $263 \div 8 = 32.875$ — a "mean without the outlier" that is lower than the mean with it, which cannot be right when the value removed was the smallest one.

10. Challenge: two complete descriptions, and a claim to judge. (Open synthesis — read the reasoning.)

Plot A. *Direction:* positive — the leftmost y is 52 and the rightmost is 88, and the points climb steadily throughout. *Form:* linear — the points follow the straight line $y = 5x + 47$ with no bend. *Strength:* strong — the residuals range only from -1 to 1, so every point sits within one unit of the line.

Plot E. *Direction:* positive — y tends to rise from 35 at the left to 71 at the right, though it zig-zags on the way. *Form:* linear — there is no bend or turning point; the trend is $y = 5x + 25$. *Strength:* weak to moderate — the residuals range from -7 to 8, so points sit as much as eight units off the line.

Judging the claim. *Disagree.* Plot E does show a positive linear association, so there is a link, but "strong" is not supported. The residual range of 15 is more than seven times Plot A's range of 2, and individual points miss the line by up to eight units — more than the whole spread of Plot A's points. A defensible sentence is: "Plot E shows a positive linear association of moderate strength; predictions from its trend line could easily be off by seven or eight."

Grading: full credit needs (i) all three words used for both plots, (ii) a number cited for each — the end values for direction, the absence of a turning point for form, the residual range for strength — and (iii) a rejection of "strong" that cites the spread rather than the slope. The commonest wrong answer agrees with the claim because Plot E's trend line rises steeply. Slope is not strength: both plots have slope 5.

Open synthesis — $\boxed{\text{open response}}$.